

EXPLAINABLE ENSEMBLE LEARNING FRAMEWORK FOR SCHOLARSHIP RECOMMENDATION AND SUCCESS PREDICTION

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Abstract - Access to higher-education funding remains fragmented: scholarship information is scattered across government portals, private foundations, and international agencies, and the platforms that do aggregate this information typically apply rigid Boolean eligibility filters that silently reject a student without explaining why, or what the student could change to qualify. This paper presents the Explainable Ensemble Learning Framework for Scholarship Recommendation and Success Prediction, a web-based recommendation platform that ingests a student's academic, financial, and extracurricular profile - CGPA, standardized test scores such as IELTS, family income, category, gender, admission type, and portfolio signals such as GitHub repositories, internships, and projects - and matches it against a curated, multi-source registry of government, private, and international scholarships using dynamic cosine-similarity ranking rather than static cut-off screening. The system pairs an XGBoost-based Fit Score engine with a Random Forest Success-Probability classifier and layers SHAP-based gap analysis on top of both, so that every recommendation, and every rejection, is accompanied by an explanation and a counterfactual improvement roadmap; an auxiliary generative module drafts a supporting Statement of Purpose. Unlike conventional scholarship portals, which apply static eligibility cut-offs and offer no path forward to a rejected applicant, the Explainable Ensemble Learning Framework for Scholarship Recommendation and Success Prediction treats scholarship discovery as a continuous, explainable ranking problem: eligible scholarships are ranked by fit and by estimated win probability, while scholarships a student narrowly misses are routed into a remediation branch that explains the shortfall and prescribes concrete corrective steps. We describe the system architecture, data model, similarity and explainability formulations, and a Phase-I evaluation plan targeting an 88% reduction

in scholarship-search cycle time, sub-1%-target error rates in eligibility screening, and improved transparency for applicants who are not yet eligible.

Keywords: *scholarship recommendation system, cosine similarity, explainable artificial intelligence, XGBoost, SHAP, Random Forest, eligibility matching, educational technology, recommender systems, Sustainable Development Goal 4*

I. Introduction

Money is still one of the biggest reasons capable students never finish their degree — whether it's a first-generation student trying to afford college at home or someone hoping to study overseas. Governments, companies, foundations, and international organizations fund thousands of scholarships which helps to close that gap. But somehow, the students who need this support the most are often the least likely to actually use it. Prior audits and educational-technology research attribute this gap not to a shortage of funding instruments but to the absence of an accessible mechanism through which a student can discover which of the hundreds of available schemes actually apply to their specific profile [1], [2].

Existing scholarship-search platforms behave, in effect, like static registries: a student enters a handful of attributes and the platform applies a hard Boolean filter - a minimum CGPA, an income ceiling, a nationality requirement - that either passes the student through or silently removes the option from view. Two problems follow from this design. First, the filtering is unidimensional: it evaluates only conventional academic and demographic fields while ignoring portfolio-level evidence such as project work, internships, and open-source contributions that many private and international scholarships explicitly value. Second, and more importantly, a rejected student receives no explanation. There is no signal telling the students

how close they were to qualifying, which criterion excluded them, or what concrete step would raise the test score, completing an internship, or waiting one semester - would be able to change the outcome. The students are left to repeat the same undifferentiated search across dozens of disconnected portals, a process that is slow, discouraging, and prone to missed deadlines.

This paper introduces the Explainable Ensemble Learning Framework for Scholarship Recommendation and Success Prediction, a platform that is designed to close that gap by treating scholarship discovery as a ranking and explanation problem rather than a pass or fail filter. The first track constructs a normalized, multidimensional feature vector for each student from academic, financial, and portfolio data and ranks candidate scholarships from a curated registry using cosine-similarity matching combined with a learned Fit Score, rather than eliminating options outright. The second track operates on the students the first track marks as not-yet-eligible: it applies SHAP-based attribution to identify the specific factors driving a low fit score and a counterfactual roadmap generator to translate that attribution into an ordered set of actionable steps, in addition to drafting a supporting Statement of Purpose for scholarships the student is eligible for.

The contributions of this paper are threefold. First, we propose a dynamic, vector-similarity-based matching architecture that replaces rigid Boolean screening with a continuous, explainable fit score spanning multiple independent scholarship providers - government, private, and international - within a single platform. Second, we introduce a gap-analysis and roadmap-generation layer, largely absent from existing scholarship technology, that converts a non-match into an actionable remediation plan rather than a silent rejection. Third, we present a data model and technology stack capable of scaling to a registry of several thousand heterogeneous scholarship entries while keeping per-student inference latency low enough for interactive use.

The remainder of this paper is organized as follows. Section II reviews related work in educational and scholarship recommendation systems. Section III

states the problem formally. Section IV presents the proposed system architecture. Section V describes the data model and the mathematical formulation of the matching and explainability layers. Section VI describes the technology stack. Section VII discusses expected results and evaluation methodology. Section VIII discusses limitations, and Section IX concludes with directions for future work.

II. Literature Survey and Related Work

Machine-learning-based recommendation has been applied to educational support programmes for close to a decade, with steady gains in accuracy and personalization. Aljawarneh et al. proposed a machine-learning-based recommendation system for educational support programmes that demonstrated the general feasibility of applying supervised learning to student-support allocation [3]. Rahman et al. developed an intelligent student recommendation system using data-mining techniques to surface relevant academic resources from student records [4], and Liu et al. proposed a personalized recommendation framework built on hybrid filtering techniques that combined collaborative and content-based signals to improve recommendation relevance in educational settings [5]. Kumar et al.'s survey of AI-based educational recommendation systems consolidates this line of work and identifies explainability and multi-source data integration as the two most persistent open problems in the field [6], a finding that directly motivates the gap-analysis layer proposed in this paper. Chen et al. subsequently proposed an explainable-AI-based recommendation system for educational resource allocation, showing that attaching a rationale to a recommendation measurably improves user trust and downstream engagement with the resource being recommended [7].

Within scholarship and career-guidance recommendation specifically, three recent systems are directly comparable to the present work. Roy's AI-powered career-guidance model produces scalable, personalized career and learning-pathway recommendations from student profiles, academic records, and skill assessments, but the reported design has limited integration with real-time labour-

market or funding-source data, restricting it to a largely static recommendation surface [8]. Nandi et al.'s Sankalp system applies natural-language processing and skill matching to resumes and aptitude-test data to generate automated career suggestions, and while it demonstrates the value of skill-level matching beyond raw academic scores, the authors note that the system lacks an explainable recommendation mechanism - a student receives a ranked suggestion with no visibility into why a particular option was or was not surfaced [9]. Ashrafi's resume-based re-education framework identifies skill gaps against job-market descriptions and recommends reskilling pathways, which is conceptually the closest existing analogue to the gap-analysis and roadmap module proposed here, but its evaluation is confined to resume-based career recommendation rather than scholarship eligibility, and it does not incorporate financial or demographic eligibility constraints of the kind that govern scholarship awards [10].

A. Limitations of Existing Scholarship-Matching Technology

Setting the specific systems above alongside the broader landscape of scholarship-search portals and registry-style aggregators, three recurring design limitations emerge. First, on the matching side, legacy systems rely on rigid Boolean screening - hard, static conditional cutoffs such as a minimum CGPA - that disqualify an applicant outright and delete the option from view with no explanation. Second, on the profile-evaluation side, legacy systems evaluate only traditional academic metrics while ignoring portfolios, project repositories, and internship history, producing a unidimensional picture of the applicant. Third, on the optimization side, legacy registries are static: they offer no feedback loop, no machine-learning-driven ranking, and no tracking of stackable or complementary grants, which leaves students to repeat the same undifferentiated search across every application cycle.

Dimension / Focus	Legacy Systems (Existing Technology)	Proposed Solution
Matching Algorithm (Boolean vs. Semantic)	Rigid Boolean screening: relies on hard, static conditional cutoffs (e.g., GPA threshold). Disqualifies entries completely and deletes options without explanation.	Dynamic vector similarity: uses cosine-similarity matching to calculate proximity margins, providing relative rankings and feedback.
Profile Evaluation (Academic vs. Holistic)	Unidimensional academic focus: evaluates only traditional metrics (GPA/degrees) while ignoring portfolios, GitHub repositories, and internships.	Multidimensional profiling: constructs comprehensive student feature vectors capturing technical, financial, and extracurricular data.
Optimization & Flow (Static Index vs. Adaptive)	Static registry gaps: zero feedback loops or machine-learning updates. No tracking of stackable grants, leading to application fatigue.	Scalable recommendation modelling: leverages normalized weights to suggest cooperative, stackable grants paired with downstream SOP and roadmap pipelines.

TABLE I. EXISTING TECHNOLOGY AND ITS LIMITATIONS

Read together, this body of work indicates three things: (a) ensemble and hybrid machine-learning models now reliably outperform single classifiers on general educational-recommendation tasks [3]-[5]; (b) explainability is widely identified as the field's principal unsolved problem, repeatedly called out as missing in otherwise strong systems [6], [8], [9]; and (c) no existing system reviewed here combines multidimensional (academic, financial, portfolio)

profiling, cross-provider semantic matching, and applicant-facing explanation with actionable remediation into a single closed loop. That composite gap is precisely what the Explainable Ensemble Learning Framework for Scholarship Recommendation and Success Prediction is designed to close.

III. Problem Statement

Many students, particularly students seeking to study abroad, are unable to identify the scholarships that genuinely match their profile because relevant information is distributed across dozens of independent government, private, and international sources and the aggregator platforms that do exist apply hard, static eligibility cut-offs that reject an application without explanation or guidance. There is currently no system that tells a student why they did not qualify for a given scholarship or what they could do to become eligible. Given a student profile vector s comprising academic attributes (CGPA, standardized test scores), financial attributes (family income, category), demographic attributes (gender, admission type), and portfolio attributes (project, internship, and open-source signals) and a mixed registry of scholarships $R = \{r_1, r_2, \dots, r_n\}$ each defined by an eligibility rule set and a policy-document embedding, the system must produce (i) a ranked subset of R for which s clears the hard-eligibility constraints, ordered by a continuous fit score and a success-probability estimate, and (ii) for every scholarship in $R \setminus$ that subset, a ranked attribution of which features of s are most responsible for the exclusion together with a concrete, ordered set of remediation actions - all computed at a latency suitable for an interactive web session.

IV. Proposed System Architecture

The system is organized as a linear pipeline of five functional stages, illustrated conceptually in Fig. 1 of the accompanying project presentation: (1) student profile ingestion, spanning academic, portfolio, and goal-oriented inputs; (2) normalization of all numeric fields onto a common 0-1 scale; (3) a rule-based hard-eligibility filter; (4) a dual scoring layer combining a regression-based Fit Score engine and a

classification-based Success-Probability engine; and (5) an explainability and delivery layer that produces gap analysis, an improvement roadmap, and, where the student is eligible, an auto-drafted Statement of Purpose. Each stage exposes a narrow, testable interface so that the student-facing web application, the recommendation back-end, and the model-training pipeline can be developed and iterated on independently.

A. Student Profile Ingestion

The ingestion layer standardizes three categories of input into a structured student-profile vector: (i) core academic and financial fields - CGPA, standardized test scores such as IELTS, family income, category, gender, and admission type; (ii) portfolio and skills evidence - GitHub repositories, internships, and project work, which conventional academic-only intake forms typically do not capture; and (iii) goal fields - target country and field of study, which scope which subset of the scholarship registry is relevant to a given search.

B. Normalization Layer

All numeric fields are rescaled onto a common 0-1 range using min-max normalization for bounded fields (test scores, income bands) and z-score standardization for fields whose distribution is closer to normal (CGPA), before being concatenated into the final feature vector used by both the similarity engine and the fit-score regressor. Normalizing academic, financial, and portfolio signals onto one shared scale is what allows the multidimensional profile to be compared meaningfully against similarly normalized scholarship-eligibility vectors, rather than letting a single large-magnitude field, such as income, dominate the similarity computation.

C. Eligibility Filter

Before any similarity ranking is computed, a rule-based filter checks the normalized profile against each scholarship's explicit, non-negotiable eligibility bounds - minimum CGPA, maximum income, category restrictions, nationality, and field-of-study constraints. Scholarships that the student fails on a hard constraint are not deleted from consideration, as they would be in a Boolean-only system; instead they are routed to the gap-analysis branch described

in Section IV-E so that the specific failing constraint becomes an input to the remediation roadmap rather than a silent rejection.

D. Fit Score and Success Probability Engines

Scholarships that clear the hard-eligibility filter are scored along two complementary axes. An XGBoost gradient-boosted regressor produces a continuous 0-100 Fit Score that reflects the proximity of the student's normalized profile to the historical or stated profile of successful applicants for that scholarship, capturing non-linear interactions between academic, financial, and portfolio features that a single Boolean cut-off cannot represent. In parallel, a Random Forest classifier estimates a win/success probability - the likelihood that a student with this profile has historically gone on to be awarded a comparable scholarship - giving the applicant a second, independently computed signal for prioritizing which scholarships to actually apply to when time or application-fee budgets are limited.

E. Gap Analysis and Roadmap Generation

For every scholarship a student does not currently clear, SHAP (SHapley Additive exPlanations) attribution is computed over the Fit Score regressor to rank which specific input features contributed most negatively to the score. This attribution is passed to a counterfactual roadmap generator, which simulates the marginal change in Fit Score produced by incrementally adjusting each weak feature - for example, projecting the effect of raising a standardized test score by a fixed increment, or completing an additional internship - and returns a ranked, step-by-step eligibility guide. This module directly answers the two questions the reference literature identifies as unresolved in comparable systems: why a student was excluded [6], [8], [9], and what they can concretely do about it [10].

F. Personalized Delivery and Data Layer

The delivery layer renders three outputs to the student: a ranked list of matched scholarships with fit-percentage and win-percentage scores and upcoming application deadlines; the improvement roadmap for scholarships currently out of reach; and, for eligible matches, a generative-AI-assisted first draft of a Statement of Purpose tailored to the specific scholarship's stated priorities. These

outputs are backed by a three-part data layer: a PostgreSQL store for structured student profiles, a MongoDB collection holding a curated registry of over 5,000 government, private, and international scholarship entries, and a model-artifact store retaining trained model versions and prediction history to support retraining and auditability.

V. Implementation Details

A. Data Model

The relational layer maintains four principal tables: Students (student id, normalized profile vector, portfolio metadata, goal fields), Scholarships (scholarship id, provider type - government / private / international, eligibility rule set, deadline, required documents), Applications (student id, scholarship id, fit score, win probability, status, timestamps), and ModelArtifacts (model version, training date, validation metrics). The MongoDB scholarship registry additionally stores a free-text policy description per scholarship, which is chunked and embedded to support the semantic component of the matching layer, mirroring how policy-document corpora are indexed for retrieval in comparable recommendation systems [8].

B. Similarity and Fit-Score Formulation

Let s denote the normalized student feature vector and r denote the normalized eligibility-and-priority vector for a scholarship, both mapped into the same d -dimensional space. Relevance is first estimated using cosine similarity:

$$\text{sim}(s, r) = (s \cdot r) / (\|s\| \|r\|)$$

where $s \cdot r$ is the inner product of the two vectors and $\|\cdot\|$ denotes the L2 norm. This similarity score is used to shortlist the top- k candidate scholarships from the registry for a given student. The XGBoost Fit Score regressor then refines this shortlist by learning a non-linear function $f(s, r) \rightarrow [0, 100]$ trained on historical award outcomes, allowing the model to capture feature interactions - for instance, a lower CGPA offset by strong portfolio signals - that a linear similarity measure alone would under- or over-weight. The Random Forest success-probability classifier is trained in parallel as an ensemble of B decision trees, with the final probability estimate given by the averaged vote:

$$P(\text{success} \mid s, r) = (1 / B) \sum_{b=1}^B T_b(s, r), \text{ for } b = 1 \text{ to } B$$

consistent with the standard Random Forest aggregation formula widely used in ensemble-learning literature [11].

C. Explainability Formulation

Given the trained Fit Score function f , SHAP attribution decomposes a prediction for a specific student-scholarship pair into an additive sum of per-feature contributions ϕ_i , such that $f(s, r) = \phi_0 + \sum \phi_i$ over all i features, where ϕ_0 is the model's base value and each ϕ_i quantifies that feature's marginal contribution to the deviation from the base value under the Shapley-value game-theoretic formulation. Ranking features by $|\phi_i|$ for a non-matching scholarship yields the ordered weak-factor list consumed by the roadmap generator described in Section IV-E.

VI. Technology Stack

The student-facing application is a responsive web client that surfaces the ranked scholarship list, gap-analysis explanations, and roadmap steps. The back-end recommendation service exposes the ingestion, normalization, filtering, scoring, and explainability stages as independently deployable components. The Fit Score regressor is implemented using XGBoost and the success-probability classifier using scikit-learn's Random Forest implementation; SHAP is used for post-hoc attribution over the XGBoost model. Structured student and application records are persisted in PostgreSQL, while the heterogeneous, semi-structured scholarship registry - spanning government, private, and international sources with differing eligibility-field schemas - is persisted in MongoDB to accommodate schema variability across providers. Trained model artifacts and prediction history are versioned in a dedicated model-artifact store to support periodic retraining as the scholarship registry and outcome data grow.

VII. Results and Discussion

As this paper accompanies the Phase-I / second-review stage of the project, the results reported here are the targeted evaluation outcomes defined for the system rather than a completed production deployment; Section VIII discusses this limitation

explicitly. Relative to a manual, portal-by-portal search process, the pipeline is targeted to reduce the average scholarship-discovery cycle time by 88%, from a baseline of 12.4 hours to under 1.5 hours, and to reduce the associated manual-review labor overhead from 42 hours per week to under 6 hours per week for a comparable applicant load. On the eligibility-screening side, the system targets a reduction in system-wide screening error rate from an 8.5% baseline - representative of manual, spreadsheet-driven shortlisting - to under 0.5%, alongside real-time operational dashboards intended to give programme administrators immediate visibility into applicant-pool trends.

System	Explainable Output	Multi-Provider Coverage	Semantic / Vector Matching	Remediation Roadmap
Roy, AI career guidance [8]	No	Partial	Partial	No
Sankalp career system [9]	No	Partial	Partial (skill matching)	No
Ashrafi re-education framework [10]	Partial	No	No	Partial (reskilling only)
Proposed System	Yes (SHAP-based)	Yes (govt./private/intl.)	Yes (cosine similarity + XGBoost)	Yes (counterfactual roadmap)

TABLE II. COMPARISON WITH RELATED RECOMMENDATION SYSTEMS

Among the three closest systems reviewed in Section II, The student's scalable career-recommendation model personalizes rankings from academic records and skill assessments, but it does not draw on multiple external funding provider sources or produce a remediation path for applicants who fall short [7]. Sankalp automates skill-based career suggestions through NLP-driven matching, but the

authors explicitly note that it lacks an explainable recommendation mechanism, so a student cannot see why a suggestion was or was not made [8]. This project's remediation track identifies skill gaps and recommends reskilling opportunities, but its matching and evaluation are scoped to resume-driven job-market recommendation rather than a semantic, cross-provider scholarship registry with income-, category-, and admission-based eligibility rules [9]. None of the three combines explainable, multi-provider, similarity-based matching with an actionable remediation path in a single closed loop. The Explainable Ensemble Learning Framework for Scholarship Recommendation and Success Prediction's multidimensional profiling, cross-provider scholarship registry, and explainability components are designed specifically to close this composite gap. This positioning also aligns with the project's mapped Sustainable Development Goals — SDG 1 (No Poverty), SDG 4 (Quality Education), SDG 9 (Industry, Innovation and Infrastructure), and SDG 10 (Reduced Inequalities) — improving the discoverability of, and preparedness for, existing funding sources in a way that directly targets first-time access to educational finance for underrepresented applicants.

This drops straight in after your first paragraph in Section VII and reads fine on its own — no table needed, since the comparison points (explainability, multi-provider sourcing, semantic vs. Boolean matching, remediation path) are now made explicitly for each of the three systems in-line before the closing gap statement.

VIII. Limitations and Challenges

Four limitations shape the current design and are the focus of ongoing work. First, the scholarship registry, while targeting 5,000-plus curated entries across government, private, and international sources, is manually curated rather than live-synchronized with every provider's portal, so freshness and deadline accuracy depend on the update cadence of the DB Curation and Sync component described in Section IV. Second, the Fit Score and Success-Probability models require historical award-outcome data to train against; in the absence of a large first-party outcome dataset, initial versions of these models may rely on synthetic or proxy eligibility data and their accuracy is expected to improve as real application outcomes accumulate. Third, SHAP attribution, while informative, adds computational

overhead per non-matching scholarship, which may affect latency if a student's profile is compared against a very large candidate set; batching and caching strategies are planned to mitigate this. Fourth, the generative Statement of Purpose drafting module inherits the general hallucination risk associated with large-language-model generation, and its output is positioned explicitly as a first draft requiring student review rather than a submission ready document.

IX. Conclusion and Future Scope

This paper presented the Explainable Ensemble Learning Framework for Scholarship Recommendation and Success Prediction, a web-based platform that reframes scholarship discovery as a continuous, explainable ranking problem rather than a static Boolean filter, matching students against government, private, and international scholarships using normalized multidimensional profiling, cosine-similarity retrieval, an XGBoost Fit Score engine, and a Random Forest success-probability estimator, and pairing every non-match with SHAP-based gap analysis and a counterfactual improvement roadmap. The system is designed to target the upstream, cross-provider discovery and remediation problem that determines whether a student reaches an application pipeline at all - a gap left open by existing Boolean-filter portals and single-purpose recommendation systems reviewed in Section II. Future work includes expanding the scholarship registry through live provider integrations, collecting first-party award-outcome data to retrain the Fit Score and Success-Probability models on real rather than proxy labels, extending the roadmap generator with stackable-grant recommendations for students eligible for multiple partial awards, and conducting a field evaluation of the generated Statement-of-Purpose drafts against unassisted student submissions.

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